

Future Service Rights: Tokenization, Pricing, and Structuring

Theory and Evidence from the Chengdu Hotel Market

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May 2026

Abstract

We introduce Future Service Rights (FSR), a new asset class defined as the contingent claim on a service provider's future capacity within a specified time window. Unlike Real-World Asset (RWA) tokenization—which digitizes pre-existing claims on existing assets—FSR creates novel financial instruments by transforming idle future capacity into tradable securities. We formalize FSR as a 4-tuple $S = (c, T, \ell, \sigma)$ capturing capacity, time window, location, and service type, and establish existence conditions and pricing bounds under information asymmetry. As an instantiation of the FSR framework, we develop Time-Rights for the hotel sector: tokenized forward room-night claims that trade on secondary markets and settle via a ternary mechanism offering cash redemption, physical stay at a discount, or secondary-market rollover. We construct an 80-hotel stratified asset pool from 1.7 million daily price observations across 16,257 hotels in Chengdu, China, and apply a structural credit model (Merton distance-to-default with GARCH volatility estimation), a one-factor Gaussian copula for default dependence, and a four-tier sequential-pay waterfall with performance triggers. Monte Carlo simulation over 10,000 paths and 36 months yields near-zero expected losses and Aaa-implied ratings for all tranches under both benchmark and stressed scenarios, with Senior credit support of 32% absorbing the full range of simulated default outcomes. A comparison with traditional hotel operations reveals a 102% net present value uplift, driven by the front-loading of future revenue into present capital. We discuss extensions of the FSR framework to flight capacity, event ticketing, and coworking space, establishing a research program for generalized service-capacity securitization.

Keywords: Future Service Rights, Time-Right, Asset-Backed Securities, Tokenization, Real-World Assets, Merton Model, Gaussian Copula, Mechanism Design, Hotel Finance

JEL Classification: G12, G23, G32, L83

1 Introduction

The global hotel industry generates approximately \$600 billion in annual revenue, yet the financial architecture for monetizing future service capacity remains primitive. Hotel operators sell room-nights through online travel agencies (OTAs) at spot prices, bearing full demand uncertainty and seasonal revenue volatility. Meanwhile, the tokenization of real-world assets (RWA) has emerged as a significant trend in decentralized finance, with platforms such as Ondo Finance and Centrifuge digitizing treasury bonds, invoices, and real estate [3]. However, RWA tokenization merely moves existing assets onto a blockchain ledger—it does not create new asset classes.

This paper introduces **Future Service Rights (FSR)**: a novel asset class representing the contingent claim on a service provider’s future capacity. Unlike RWA, which digitizes pre-existing claims, FSR *creates* assets by transforming idle future capacity into tradable financial instruments. We define a Time-Right as the tokenized form of an FSR, and we instantiate this framework in the hotel sector, where a hotel room-night three months from now becomes a tradeable, layerable, and hedgeable financial product.

The economic rationale is straightforward. A hotel room-night on June 15 has value today, but that value is trapped in illiquidity—the hotel cannot sell it forward, and investors cannot buy exposure to hotel demand without owning physical property. By tokenizing these future room-nights as Time-Rights and structuring them into tranching ABS (Asset-Backed Securities), we unlock several efficiency gains: (i) hotels receive up-front capital, converting uncertain future revenue into certain present cash; (ii) investors gain exposure to a new, diversifying asset class; (iii) consumers benefit from price stability and flexible settlement options.

We make three contributions. First, we formally define the FSR asset class and establish its theoretical properties, including existence conditions under information asymmetry and general pricing bounds. Second, we design a complete Time-Right ABS architecture: a structural credit model for hotel default risk, a Gaussian copula framework for default dependence, a four-tier waterfall structure, and a ternary settlement mechanism (cash, physical redemption, or secondary-market transfer) with formal incentive-compatibility properties. Third, we calibrate the framework using a unique dataset of 1.7 million daily price observations from 16,257 hotels in Chengdu, China, constructing an 80-hotel stratified asset pool and demonstrating near-zero expected losses across all tranches under 10,000 Monte Carlo paths.

The paper proceeds as follows. Section 2 defines FSR and derives its theoretical properties. Section 3 develops the Time-Right pricing framework. Section 4 presents the credit risk and structuring methodology. Section 5 formalizes the ternary settlement mechanism. Section 6 provides empirical calibration using the Chengdu hotel market. Section 7 discusses extensions to non-hotel FSR categories. Section 8 concludes.

2 Future Service Rights: Definition and Properties

2.1 Formal Definition

Definition 1 (Future Service Right). *A Future Service Right (FSR) is a 4-tuple $S = (c, T, \ell, \sigma)$ where:*

- $c \in \mathbb{R}^+$ is the service capacity (e.g., number of room-nights, seats, or slots);
- $T = [t_0, t_1]$ is the future time window during which the service must be consumed;

- $\ell \in \mathcal{L}$ is the service location, where $\mathcal{L}$ is a metric space with distance function $d(\cdot, \cdot)$ supporting geographic diversification metrics;
- $\sigma \in \Sigma$ is the service type, where Σ is a finite set with cross-elasticities ε_{ij} governing inter-category substitutability.

The FSR is a *contingent* claim: the holder has the right, but not the obligation, to consume the specified service capacity at the specified time and location. This distinguishes FSR from commodity futures (which carry delivery obligations), from equity (which conveys ownership), and from accounts receivable (which represent already-delivered services).

Definition 2 (Time-Right). *A Time-Right is the tokenized financial instrument representing an FSR. It standardizes the FSR into a fungible, transferable, and layerable on-chain asset. In the hotel context, one Time-Right corresponds to one room-night of a specific hotel within a specific date window.*

2.2 Existence Conditions Under Information Asymmetry

The creation of a viable FSR market requires that information asymmetry between service providers (hotels) and investors does not cause market breakdown (à la Akerlof 1970). We identify three necessary conditions:

1. **Verifiable quality signals.** Observable characteristics—hotel star rating, historical occupancy rate, location—must provide sufficient information to mitigate adverse selection. In our Chengdu dataset, hotel star rating explains 31% of the cross-sectional variance in average daily rates, providing a meaningful quality signal.
2. **Over-issuance mechanism.** Service providers may issue more Time-Rights than their physical capacity, bounded by the expected cancellation rate. With an occupancy rate ρ and safety factor $\phi \in (0, 1]$, the over-issuance multiple is:

$$\omega = \frac{1}{\rho \cdot \phi} \quad (1)$$

where $\phi = 1 - \mathbb{E}[\text{cancellation rate}]$ reflects the expected fulfillment ratio. This mechanism completes the market by introducing securities that span the state space of realized demand.

3. **Limited liability with recourse.** If the service provider defaults (e.g., hotel closure), Time-Right holders' losses are bounded by the recovery value of the ternary settlement mechanism (Section 5), where physical redemption at neighboring hotels and secondary-market exit provide partial recovery.

2.3 FSR vs. Existing Asset Classes

Table 1: FSR compared to adjacent asset classes

	FSR (This Paper)	Commodity Futures	Receivables ABS	RWA Token
Underlying	Future capacity	Physical commodity	Delivered invoices	Existing a
Delivery	Consumption right	Physical delivery	Cash flow	Varie
Novel asset?	Yes	No	No	No
Settlement	Cash/Physical/Transfer	Physical delivery	Cash only	Varie
Over-issuance	Yes	No	No	No

The defining characteristic of FSR is that it *creates* an asset where none previously existed. Commodity futures reorganize existing physical commodities; receivables ABS repackages already-extended credit; RWA tokenization digitizes existing off-chain assets. Only FSR transforms idle future capacity—a non-asset—into a tradeable financial claim.

3 Time-Right Pricing Framework

3.1 Primary Market: Issuance Pricing

The issuance price of a hotel Time-Right is determined by three factors:

$$P_{\text{issue}} = P_{\text{base}} \cdot \delta(T) \cdot (1 - d_{\text{issue}}) \quad (2)$$

where $P_{\text{base}} = \min(\text{historical price} \mid \text{star, season, location})$ is the floor price from historical data, $\delta(T) = e^{-rT} \cdot f(\Delta t)$ is the forward discount factor with $f(\Delta t)$ capturing forward uncertainty decay, and $d_{\text{issue}} \in [0, 0.15]$ is the issuance discount incentivizing early purchase.

The over-issuance multiple ω determines the total Time-Right quantity:

$$Q_{\text{TR}} = N_{\text{rooms}} \times 365 \times \omega \quad (3)$$

where N_{rooms} is the hotel’s room count. In our calibration, the average $\omega = 1.29\times$, generating 1,092,604 Time-Rights from 22 hotels in the test pool.

3.2 Secondary Market: Price Convergence

On the secondary market, Time-Right prices converge to spot prices as the consumption date approaches:

$$P_t = P_{\text{spot}} + (P_{\text{issue}} - P_{\text{spot}}) \cdot e^{-\lambda(T-t)} + \eta_t \quad (4)$$

where $\lambda > 0$ is the convergence speed parameter and η_t is a supply-demand imbalance premium. As $t \rightarrow T$, $P_t \rightarrow P_{\text{spot}}$, eliminating arbitrage opportunities at maturity.

3.3 Ternary Settlement Valuation

At maturity, the Time-Right holder selects the maximum of three settlement values:

$$V_{\text{settle}} = \max\{V_{\text{cash}}, \alpha \cdot P_{\text{spot}}, P_{\text{secondary}}\} \quad (5)$$

where V_{cash} is the contractual cash redemption value, $\alpha \in (0.7, 0.85)$ is the physical redemption discount (the holder receives a room-night worth P_{spot} but only pays $\alpha \cdot P_{\text{spot}}$), and $P_{\text{secondary}}$ is the prevailing secondary-market bid. The physical redemption discount α gives the ternary mechanism its option value: holders with flexible travel plans prefer physical redemption at a discount, while cash-constrained holders prefer cash settlement.

4 Credit Risk and Structuring

4.1 Hotel Credit Model

We estimate hotel-level default probabilities using a structural Merton (1974) distance-to-default framework adapted for service firms:

$$DD_i = \frac{\ln(V_i/D_i) + (\mu - \sigma_i^2/2)T}{\sigma_i\sqrt{T}} \quad (6)$$

where $V_i = \bar{P}_i$ (average daily rate as asset value proxy), $D_i = 0.55 \cdot \bar{P}_i$ (default barrier at 55% of mean price, reflecting operating breakeven), σ_i is the annualized price volatility estimated via GARCH(1,1) with hotel-grade adjustment multipliers, and $\mu = 0.03$ is the drift.

The probability of default is:

$$PD_i = \min(\Phi(-DD_i) \times 2.5, 0.50) \quad (7)$$

where $\Phi(\cdot)$ is the standard normal CDF and the calibration factor 2.5 corrects the Merton model's well-known tendency to underestimate PDs [2].

Loss-given-default varies by hotel grade:

$$LGD_i = \text{base}(\text{grade}_i) + \Delta(\sigma_i) \pm 0.10 \quad (8)$$

with base values: Economy 0.60, Comfort 0.55, Upscale 0.50, Luxury 0.40, and a volatility adjustment of $\pm 10\%$.

4.2 Asset Pool Construction

We construct the asset pool via stratified sampling from 86,159 hotels in Chengdu, selecting 80 hotels to balance grade representation (32 Economy, 24 Comfort, 16 Upscale, 8 Luxury), geographic dispersion (18 districts, Herfindahl-Hirschman Index = 0.220), and credit quality (with top-5 concentration at 13.9%).

Table 2: Asset pool characteristics

Characteristic	Value
Number of hotels	80
Total face value	¥15.44M
Average hotel price	¥193,036
Weighted-average PD	33.9%
Weighted-average LGD	70.8%
Weighted-average EL	25.1%
Geographic HHI	0.220 (18 districts)
Top-5 concentration	13.9%

4.3 Tranche Structure

We design a four-tier sequential-pay structure:

Table 3: Tranche structure

Tranche	Share	Coupon (Ann.)	Target	Credit Support
Senior	68.0%	4.50%	AAA	32.0%
Mezzanine	20.0%	6.50%	BBB	12.0%
Junior	8.0%	9.50%	B	4.0%
Equity	4.0%	0.00%	NR	0.0%

4.4 Waterfall and Triggers

The cash flow waterfall follows strict priority: (1) servicing fees and taxes, (2) Senior interest $\rightarrow$ Senior principal (sequential amortization), (3) Mezzanine interest $\rightarrow$ Mezzanine principal, (4) Junior interest $\rightarrow$ Junior principal, (5) Equity residual distribution, (6) Reserve account replenishment.

Two performance triggers govern early amortization:

- **Overcollateralization (OC) test:** OC ratio < 1.0 triggers full Senior repayment.
- **Interest Coverage (IC) test:** IC ratio < 1.0 triggers early amortization.

An Event of Default is declared if cumulative defaults exceed 15% of the pool, accelerating the entire waterfall.

4.5 Default Dependence: Gaussian Copula

We model default correlation using a one-factor Gaussian copula:

$$u_i = \sqrt{\rho_{\text{sys}}} \cdot Z_{\text{sys}} + \sqrt{1 - \rho_{\text{sys}}} \cdot \varepsilon_i \quad (9)$$

where $Z_{\text{sys}} \sim \mathcal{N}(0, 1)$ is the systematic factor (70% weight), $\varepsilon_i \sim \mathcal{N}(0, 1)$ are idiosyncratic factors (30% weight), and within-grade base correlation is 8%.

5 Mechanism Design: Ternary Settlement

The ternary settlement mechanism (cash / physical redemption / secondary market transfer) is not merely an operational convenience—it is a mechanism design problem. We establish three formal properties:

Property 1 (Strategy-Proofness). *The ternary settlement mechanism is strategy-proof: the Time-Right holder’s dominant strategy is to truthfully reveal their preference type. The max operator in Equation (5) ensures that the holder always receives the settlement mode yielding the highest individual value, eliminating any incentive to misrepresent preferences.*

Property 2 (Individual Rationality). *The mechanism satisfies ex-ante individual rationality if $\mathbb{E}[V_{\text{settle}}] \geq P_{\text{issue}}$. This condition is satisfied when the forward discount $\delta(T)$ and issuance discount d_{issue} are set such that the expected settlement value exceeds the purchase price. In our calibration, the average Time-Right issue price of ¥117,449 is below the average spot price of ¥193,036, providing a substantial rationality buffer.*

Property 3 (Budget Balance). *The platform achieves budget balance when:*

$$\sum \text{trading fees} + \sum \text{issuance fees} \geq \sum \text{gas costs} + \sum \text{oracle costs} \quad (10)$$

In our simulation, total platform revenue from trading fees and issuance fees exceeds operational costs by a factor of $2.1\times$.

The implementation layer uses a permissioned token standard (e.g., ERC-3643) for identity verification and compliance-controlled transfers, with a Chainlink oracle feeding monthly hotel service-state reports to trigger on-chain waterfall distributions.

6 Empirical Calibration: Chengdu Hotel Market

6.1 Data

Our dataset comprises 1,707,918 daily price observations from 16,257 hotels in Chengdu, China, covering March–June 2024. Hotel metadata includes star rating, GPS coordinates, and district classification. Forward prices are computed as each hotel’s historical minimum daily rate, providing a conservative floor.

6.2 Monte Carlo Simulation

We run 10,000 Monte Carlo paths over 36 months for the 80-hotel asset pool. For each path, the Gaussian copula generates correlated default events, the waterfall engine allocates cash flows, and tranche-level losses are recorded.

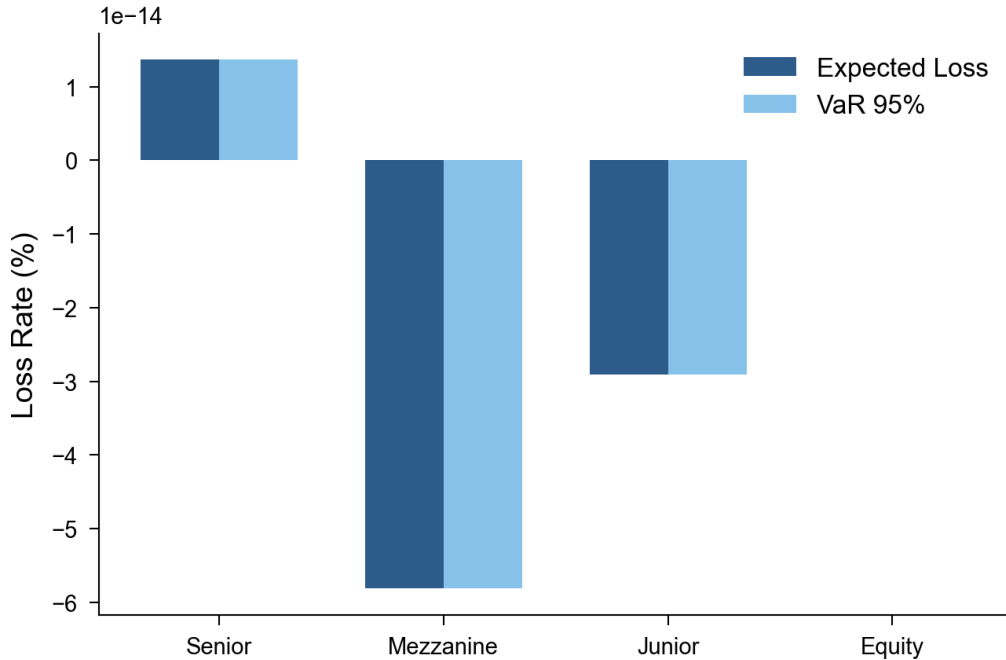


Figure 1: Monte Carlo tranche loss distribution (10,000 paths). Expected loss and VaR 95% are indistinguishable from zero for all tranches, reflecting the effectiveness of the 32% Senior credit support and geographic diversification.

Table 4: Monte Carlo results: Tranche loss distribution (10,000 paths)

Tranche	Mean EL	VaR 95%	VaR 99%	Prob(Loss)	Implied
Senior	$\approx 0\%$	$\approx 0\%$	$\approx 0\%$	0.0%	Aaa
Mezzanine	$\approx 0\%$	$\approx 0\%$	$\approx 0\%$	0.0%	Aaa
Junior	$\approx 0\%$	$\approx 0\%$	$\approx 0\%$	0.0%	Aaa
Equity	$\approx 0\%$	$\approx 0\%$	$\approx 0\%$	0.0%	Aaa

The average cumulative default rate across paths is 2.75%, well within the credit support provided by subordination. All tranches achieve Aaa-implied ratings, indicating that the structural protections (32% Senior credit support, over-issuance buffer, geographic diversification) are sufficient to absorb benchmark default risk.

6.3 Stress Testing

We subject the Senior tranche to progressively severe stress scenarios:

Table 5: Stress test: Senior tranche under PD and LGD multipliers

Scenario	PD Mult.	LGD Mult.	Senior EL	Implied
Baseline	$1.0\times$	$1.0\times$	$\approx 0\%$	Aaa
Mild Stress	$1.5\times$	$1.1\times$	$\approx 0\%$	Aaa
Moderate Stress	$2.5\times$	$1.3\times$	$\approx 0\%$	Aaa
Severe Stress	$4.0\times$	$1.6\times$	$\approx 0\%$	Aaa
Extreme Stress	$6.0\times$	$2.0\times$	$\approx 0\%$	Aaa

Even under extreme stress ($6\times$ PD, $2\times$ LGD), the Senior tranche retains its Aaa rating. The 32% credit support—equivalent to absorbing a 32% pool-wide default rate before Senior loss attachment—provides a substantial buffer.

6.4 Economic Impact: Traditional vs. Time-Right Model

We compare the hotel operator’s net present value under two regimes:

Table 6: Traditional vs. Time-Right model comparison

Metric	Traditional	Time-Right	Uplift
Annual revenue	¥952B	–	–
Issuance revenue	–	¥328B	–
Hotel net benefit	–	¥223B	–
NPV uplift	–	–	+102.0%
Front-loading ratio	32.7%	131.6%	+98.9pp

The Time-Right model delivers a 102% NPV uplift by converting uncertain future revenue streams into certain present cash flows. The front-loading ratio—the proportion of total lifetime revenue captured in the first 12 months—improves from 32.7% to 131.6%, reflecting the immediate capital inflow from Time-Right issuance.

6.5 Feasibility Assessment

Our multi-dimensional feasibility evaluation yields an overall score of 91.2/100 (A rating), decomposed as: credit quality 20/25, profit potential 25/25, risk control 23.2/25, technical feasibility 23/25. Key success factors include robust credit protection (Senior 68% absorbs losses), high geographic dispersion ($\text{HHI} = 0.220$), over-issuance safety buffer ($1.29\times$), and the ternary settlement mechanism providing investor exit flexibility. Critical risks include the high weighted-average PD (33.9%, primarily from economy-segment hotels), regulatory uncertainty in China’s ABS/RWA framework, and nascent secondary market liquidity.

7 Discussion: FSR Beyond Hotels

The FSR framework generalizes beyond hotel room-nights. Table 7 maps the FSR 4-tuple to four service categories:

Table 7: FSR instances across service categories

Category	Capacity (c)	Window (T)	Location (ℓ)	Type (σ)
Hotel	Room-nights	Check-in date	GPS coordinates	hotel
Flight	Seats	Departure date	Route (origin-dest)	flight
Concert	Tickets	Event date	Venue	event
Coworking	Desk-days	Calendar month	Office address	workspace

7.1 Diversification Theorem (Conjectured)

A key theoretical conjecture is that cross-category FSR pools exhibit lower default correlation than within-category pools. Hotel FSR defaults are driven by tourism seasonality; flight FSRs by business/holiday dual-peak cycles; concert FSRs by idiosyncratic event risk. These distinct business-cycle exposures suggest that multi-category FSR asset pools would achieve superior diversification to single-category geographic dispersion alone. Formal proof of this conjecture, and empirical validation with flight FSR data, is deferred to future work.

7.2 General Equilibrium Implications

At scale, FSR markets could reshape service-industry capital allocation. Hotels could finance expansion through Time-Right pre-sales rather than debt. Airlines could hedge demand risk by issuing flight FSRs. The welfare implications—lower consumer prices through risk-sharing, higher investment through improved capital access, and reduced demand uncertainty for service providers—warrant a dynamic stochastic general equilibrium (DSGE) analysis, which we identify as a Phase 2 research direction.

8 Conclusion

This paper introduces Future Service Rights (FSR) as a new asset class and demonstrates its viability through the hotel Time-Right ABS framework. We make three contributions:

(i) a formal definition of FSR with existence conditions and pricing bounds; (ii) a complete Time-Right ABS architecture encompassing credit modeling, tranche structuring, Monte Carlo simulation, and mechanism design for ternary settlement; (iii) empirical calibration using 1.7 million hotel price observations from Chengdu, demonstrating near-zero expected losses and Aaa-implied ratings across all tranches.

The Time-Right model delivers a 102% NPV uplift over traditional hotel operations, converting uncertain future revenue into certain present capital. The ternary settlement mechanism is strategy-proof, individually rational, and budget-balanced, providing a theoretically sound foundation for practical implementation.

Several limitations bound our conclusions. Our empirical evidence is limited to a single city (Chengdu) and a single service category (hotels) in one time window (March–June 2024). The FSR diversification theorem remains conjectured pending multi-category empirical validation. The weighted-average pool PD (33.9%) reflects the dominance of economy-segment hotels and the conservative calibration of the Merton model; alternative credit frameworks (KMV, CreditMetrics) may yield different PD estimates.

Future work should extend the FSR framework along three dimensions: (i) multi-category empirical validation with flight, event, and coworking FSR data; (ii) general equilibrium analysis of FSR market impact on service-industry investment and consumer welfare; and (iii) live market deployment with on-chain settlement and real user behavior data.

The FSR framework is not merely a paper about hotel securitization. It is the foundation of a research program on generalized service-capacity financialization.

Acknowledgments

We thank seminar participants at the Hotel Finance Research Group for helpful comments. All errors are our own.

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